

GWR_basic_cheetsheet

GWR model

$$y_i = \beta_0(u_i, v_i) + \beta_1(u_i, v_i)x_{i1} + \beta_2(u_i, v_i) + \dots + \epsilon_i$$

where (u_i, v_i) : 迴歸點 i , $\beta_k(u_i, v_i)$: β_k 在點 i 的值

1. 矩陣向量寫法

$$\mathbf{y} = (\mathbf{B} \otimes \mathbf{X}) \mathbf{1} + \boldsymbol{\epsilon}$$

$$\text{where } \mathbf{B} = \begin{bmatrix} \beta_0(u_1, v_1) & \beta_1(u_1, v_1) & \cdots & \beta_k(u_1, v_1) \\ \beta_0(u_2, v_2) & \beta_1(u_2, v_2) & \cdots & \beta_k(u_2, v_2) \\ \vdots & \vdots & \ddots & \vdots \\ \beta_0(u_n, v_n) & \beta_1(u_n, v_n) & \cdots & \beta_k(u_n, v_n) \end{bmatrix}, \mathbf{1} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$$

2. $\hat{\beta}$ 係數解

$$\hat{\beta}(u_i, v_i) = [\mathbf{X}^T \mathbf{W}(u_i, v_i) \mathbf{X}]^{-1} \mathbf{X}^T \mathbf{W}(u_i, v_i) \mathbf{y}$$

$$\text{where } \mathbf{W}(u_i, v_i) = \begin{bmatrix} w_{i1} & 0 & \cdots & 0 \\ 0 & w_{i2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & w_{in} \end{bmatrix} : \text{迴歸點 } i \text{ 權重矩陣}$$

比較一般迴歸係數解:

$$\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

3. 局部標準誤

$$\text{Let } \hat{\beta}(u_i, v_i) = [\mathbf{X}^T \mathbf{W}(u_i, v_i) \mathbf{X}]^{-1} \mathbf{X}^T \mathbf{W}(u_i, v_i) \mathbf{y} = \mathbf{C} \mathbf{y}$$

$$\Rightarrow V(\hat{\beta}(u_i, v_i)) = \mathbf{C} \mathbf{C}^T \sigma^2, SE(\hat{\beta}(u_i, v_i)) = \sqrt{V(\hat{\beta}(u_i, v_i))}$$

$$\text{Where } \sigma^2 = \frac{\sum (y_i - \hat{y}_i)^2}{n - 2v_1 + v_2}, v_1 = \text{tr}(\mathbf{S}), v_2 = \text{tr}(\mathbf{S}^T \mathbf{S}), \mathbf{S} : \text{hat matrix}(\hat{\mathbf{y}} = \mathbf{S} \mathbf{y})$$

$$\mathbf{S}_{(i)} = \mathbf{X}_{(i)} [\mathbf{X}^T \mathbf{W}(u_i, v_i) \mathbf{X}]^{-1} \mathbf{X}^T \mathbf{W}(u_i, v_i)$$

4. 權重選擇: bi-square

$$w_{ij} = \begin{cases} [1 - (\frac{d_{ij}}{b})^2]^2, & d_{ij} < b \\ 0, & \text{o. w.} \end{cases}$$

5. 可變(adaptive)帶寬 : N近鄰帶寬 + bi-square

$$w_{ij} = \begin{cases} [1 - (\frac{d_{ij}}{b})^2]^2, & j \text{ 是離迴歸點 } i \text{ 最近的 } N \text{ 個點之一} \\ 0, & \text{o. w.} \end{cases}$$

6. 帶寬選擇方法:

1. 交叉驗證 CV

$$CV = \sum_{i=1}^n [y_i - \hat{y}_{\neq i}(b)]^2$$

2. 廣義交叉驗證 GCV

$$GCV = n \sum_{i=1}^n [y_i - \hat{y}_i(b)]^2 / (n - v_1)^2, \text{ where } v_1 = \text{tr}(\text{hat matrix } S)$$

3. AICc

$$AIC_c = 2n \log(\hat{\sigma}) + n \log(2\pi) + n \left[\frac{n + \text{tr}(S)}{n - 2 - \text{tr}(S)} \right] \text{ where, } \sigma^2 = \frac{RSS}{n}$$

4. BIC

$$BIC = -2 \log(L) + k \log(n), \text{ where } L: \text{ model likelihood}$$